```c
#include <stdio.h>
#include <stdlib.h>
#include <pthread.h>

#define NUM_THREADS 5
#define ITERATIONS 100000

// Shared counter variables
long shared_counter_unsafe = 0;
long shared_counter_safe = 0;

// Mutex lock for synchronization
pthread_mutex_t lock;

// Thread function WITHOUT synchronization (Demonstrates Race >
void *increment_unsafe(void *arg) {
    for (int i = 0; i < ITERATIONS; i++) {
        shared_counter_unsafe++; // Race condition occurs here
    }
    return NULL;
}

// Thread function WITH synchronization (Using Mutex Locks)
void *increment_safe(void *arg) {
    for (int i = 0; i < ITERATIONS; i++) {
        pthread_mutex_lock(&lock);
        shared_counter_safe++; // Safe critical section
        pthread_mutex_unlock(&lock);
    }
    return NULL;
}

int main() {
```

```
 GNU nano 8.7.1          multithread_counter.c *
    }
    // Wait for all threads to finish
    for (int i = 0; i < NUM_THREADS; i++) {
        pthread_join(threads[i], NULL);
    }

    long expected_value = (long)NUM_THREADS * ITERATIONS;
    printf("Expected Counter Value : %ld\n", expected_value);
    printf("Actual Unsafe Counter  : %ld (Mismatch due to race>


    printf("=== 2. Demonstrating Synchronization (With Mutex L>
    // Initialize mutex
    pthread_mutex_init(&lock, NULL);

    // Create threads for safe counter
    for (int i = 0; i < NUM_THREADS; i++) {
        pthread_create(&threads[i], NULL, increment_safe, NULL>
    }
    // Wait for all threads to finish
    for (int i = 0; i < NUM_THREADS; i++) {
        pthread_join(threads[i], NULL);
    }

    printf("Expected Counter Value : %ld\n", expected_value);
    printf("Actual Safe Counter    : %ld (Matches perfectly du>

    // Destroy mutex
    pthread_mutex_destroy(&lock);

    return 0;
}
```

```
PS C:\Users\Gireesh> wsl
gireesh@Kannigantiii:/mnt/c/Users/Gireesh$ cd ~/oslab
gireesh@Kannigantiii:~/oslab$ nano multithread_counter.c
gireesh@Kannigantiii:~/oslab$ gcc -Wall multithread_counter.c -
lpthread -o multithread_counter
gireesh@Kannigantiii:~/oslab$ ./multithread_counter
```

```
gireesh@Kannigantiii:~/oslab$ ./multithread_counter
=== 1. Demonstrating Race Condition (Without Mutex) ===
Expected Counter Value : 500000
Actual Unsafe Counter  : 281081 (Mismatch due to race condition
!)

=== 2. Demonstrating Synchronization (With Mutex Locks) ===
Expected Counter Value : 500000
Actual Safe Counter    : 500000 (Matches perfectly due to mutex
 locks!)
gireesh@Kannigantiii:~/oslab$
```